37. What is Cloud Controller Manager?

Professional Answer

Cloud Controller Manager integrates Kubernetes with cloud-provider APIs for managing load balancers, volumes, and node lifecycle.

Unprofessional Answer

It manages cloud things.

Why is cloud-controller-manager used in Kubernetes?
cloud-controller-manager integrates Kubernetes with cloud provider services.
It manages cloud-specific resources like load balancers, volumes, and node lifecycle.
This separation keeps Kubernetes core cloud-agnostic.
It communicates directly with cloud provider APIs such as AWS, GCP, or Azure.
It allows Kubernetes to run consistently across different cloud environments.

. cloud-controller-manager (Cloud Internals)

Why it exists

- Separates cloud logic from core Kubernetes

Internal Controllers

1. **Node Controller**
    - Cloud node lifecycle
2. **Service Controller**
    - Creates cloud load balancers
3. **Route Controller**
    - Manages network routes
4. **Volume Controller**
    - Attaches/detaches disks

Works with

- AWS APIs
- GCP APIs
- Azure APIs

Nice choice 😄 — **cloud-controller-manager (CCM)** is one of those components that **separates average DevOps engineers from senior ones** in interviews.

Let's go **deep, real-world, and interview-ready**.

☁️ **Cloud Controller Manager (CCM) – Kubernetes + Cloud Glue**

1. What is Cloud Controller Manager?

Cloud Controller Manager is a **Kubernetes control-plane component** that integrates Kubernetes with **cloud provider APIs**.
It allows Kubernetes to manage cloud resources **without hard-coding cloud logic into core components**.
CCM runs cloud-specific controllers.
It communicates with both **kube-apiserver** and **cloud APIs**.
Without CCM, Kubernetes cannot manage cloud-native resources properly.

2. Why Cloud Controller Manager was introduced

Earlier, cloud logic was inside:

- kube-controller-manager
- kubelet

This caused:

- Tight cloud coupling
- Hard upgrades
- Security risks

CCM **decouples cloud logic** from Kubernetes core.

👉 This is a **design evolution question** interviewers love.

3. Where does CCM run?

- As a **control plane component**
- Usually as a **pod** (static pod or deployment)
- Runs only in **cloud environments**
- Not needed for bare-metal clusters

Managed services (EKS, GKE, AKS) run it for you.

4. Core controllers inside Cloud Controller Manager

1 Node Controller (Cloud-aware)

- Updates Node objects with cloud metadata

- Handles cloud node deletion
- Manages node lifecycle when VM is deleted

Example:

If an EC2 instance is terminated, CCM removes the node from Kubernetes.

2 Route Controller

- Manages cloud network routes
- Required in some cloud CNI setups
- Ensures pod-to-pod routing across nodes

Mostly used in older networking models.

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3 Service Controller (Load Balancer)

- Creates cloud LoadBalancers
- Maps Service type=LoadBalancer
- Handles lifecycle of LBs

Example:

Service → AWS ELB / GCP LB / Azure LB

4 Volume Controller

- Manages cloud volumes
- Attaches/detaches disks
- Works with CSI drivers

Example:

EBS disk attached to Pod via PVC

5. How CCM works (request flow)

1. User creates Service type LoadBalancer
2. API Server stores object
3. CCM watches Service objects
4. CCM calls cloud provider API
5. Load Balancer created
6. Service updated with external IP

This **watch–reconcile loop** is key.

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6. What happens if CCM is down?

Feature	Impact
Existing Pods	Continue running
New LoadBalancers	Not created
Node deletion	Not detected
Volume attach	May fail
Cloud sync	Broken

👉 Cluster works **partially**, not fully.

7. CCM vs kube-controller-manager

kube-controller-manager	cloud-controller-manager
Core controllers	Cloud-specific controllers
Cluster logic	Cloud infra logic
Always needed	Cloud-only
No cloud API calls	Talks to cloud APIs

8. Real production outage example

Incident

- CCM crashed after IAM permission change

Symptoms

- LoadBalancers stuck in Pending
- Node deletion not reflected
- PVC attach failures

Root cause

CCM lacked permission to call cloud APIs

Fix

- Restored IAM roles
- Restarted CCM pods

9. Common CCM misconfigurations

- Missing cloud credentials
- Incorrect IAM roles
- Wrong cloud region
- Disabled required controllers
- Network access blocked

10. Interview question: Is CCM mandatory?

Answer:
Mandatory in cloud environments.
Not required for on-prem or bare metal.
Managed Kubernetes runs it implicitly.
Without it, cloud integrations fail.

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11. How CCM communicates securely

- Uses **ServiceAccount**
- Uses cloud IAM roles
- TLS with API Server
- Restricted permissions (least privilege)

12. Ultra-hard interview question

Q:
Why is Service type LoadBalancer not created if CCM is down but Ingress works?

Answer:
Ingress may rely on existing LBs.
Service type LoadBalancer requires CCM to create cloud LBs.
Ingress controllers may already have infra.
CCM handles LB lifecycle, not routing logic.

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13. Professional vs Unprofessional answer

Unprofessional:

CCM connects Kubernetes to cloud.

Professional:

Cloud Controller Manager is a control-plane component that reconciles Kubernetes objects with cloud provider resources, enabling dynamic management of nodes, load balancers, networking, and volumes through cloud APIs.

14. Exam one-liner

Cloud Controller Manager enables Kubernetes to interact with cloud infrastructure by managing cloud-specific resources.

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15. Senior-level red flags

🚩 "CCM schedules Pods"
🚩 "CCM replaces kubelet"
🚩 "CCM stores data"

(All wrong.)

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16. Golden rule

If cloud resources don't match Kubernetes state, CCM is the suspect.